HW7

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Problem 1

Although a neural network that predicts the language of origin for last names can be useful, it poses risks if applied. Such a system could reinforce stereotypes or allow for discrimination in hiring or law enforcement by making assumptions about individuals based solely on their names. Moreover, the model is prone to errors, and misclassification could lead to biased decision making.

Problem 2

(b)

(i) Let $E := X - W_2 W_1 X$. Using standard matrix–calculus identities $\nabla_W \|A - BW\|_F^2 = -2B^\top (A - BW)$ and $\nabla_W \|A - WB\|_F^2 = -2(A - WB)B^\top$, we obtain

$$\nabla_{W_2} \mathcal{L} = -2\,(X - W_2 W_1 X)(X^\top W_1^\top), \qquad \nabla_{W_1} \mathcal{L} = -2\, W_2^\top (X - W_2 W_1 X) X^\top.$$

Hence any minimizer $(W_1^\star, W_2^\star)$ must satisfy

$$(X - W_2^\star W_1^\star X)\, X^\top W_1^{\star\top} = 0, \tag{2}$$

$$W_2^{\star\top} (X - W_2^\star W_1^\star X)\, X^\top = 0. \tag{1}$$

(ii) Let $XX^\top = U \Sigma U^\top$ with $\Sigma = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_m^2)$ and let $U_k \in \mathbb{R}^{m \times k}$ be the top-k eigenvectors. Set

$$W_1^\star = U_k^\top, \qquad W_2^\star = U_k \qquad \Rightarrow \qquad W_2^\star W_1^\star = U_k U_k^\top =: P_k.$$

Then, using $XX^\top U_k = U_k \Sigma_k$ and $(I - P_k) U_k = 0$,

$$(X - P_k X) X^\top U_k = (I - P_k)\, XX^\top U_k = (I - P_k)\, U_k \Sigma_k = 0,$$

so (2) holds. Similarly,

$$U_k^\top (X - P_k X) X^\top = U_k^\top XX^\top - U_k^\top P_k XX^\top = \Sigma_k U_k^\top - \Sigma_k U_k^\top = 0,$$

so (1) holds. Thus the principal subspace solution $(W_1, W_2) = (U_k^\top, U_k)$ satisfies the first–order conditions.

Problem 3

(a)

(i) 2 (one linear layer for the encoder W_1 and oanother for the decoder W_2, no activations).

(ii) `nn.MSELoss` (the data term is a squared Frobenius/MSE reconstruction error).

(iii) Weight Decay and SGD optimizer

(b)

Write SVDs $W_1 = U_1 \Sigma_1 V_1^\top$ and $W_2 = U_2 \Sigma_2 V_2^\top$. The regularizer equals $\lambda\big(\|W_1\|_F^2 + \|W_2\|_F^2\big) = \lambda\big(\|\Sigma_1\|_F^2 + \|\Sigma_2\|_F^2\big)$, which depends only on the singular values.

For small $\lambda > 0$, the reconstruction term is minimized when $W_2 W_1$ closely matches the rank-k projector $P_k = U_k U_k^\top$ onto the principal subspace. Any factorization of a rank-k projector admits k nonzero singular directions in W_1 and W_2 with products $\sigma_i(W_2)\,\sigma_i(W_1) \approx 1$ (along the principal directions). Minimizing

$$\sum_{i=1}^{k} \big(\sigma_i(W_1)^2 + \sigma_i(W_2)^2\big) \quad \text{subject to} \quad \sigma_i(W_1)\,\sigma_i(W_2) \approx 1$$

decouples across i; each term is minimized at $\sigma_i(W_1) = \sigma_i(W_2) = 1$ (since for $a, b > 0$ with $ab = 1$, $a^2 + b^2 \geq 2$ with equality at $a = b = 1$). Thus the penalty favors singular values equal to 1 along the k active directions and 0 otherwise. Having all nonzero singular values equal to 1 implies W_2 has orthonormal columns and W_1 has orthonormal rows. Hence, a small nonzero λ introduces an inductive bias toward W_2 with approximately orthonormal columns (and W_1 with approximately orthonormal rows) while fitting the principal subspace.

Uniqueness holds up to a $k \times k$ orthogonal change of basis: $(W_2, W_1) = (U_k Q,\ Q^\top U_k^\top)$ for any $Q \in O(k)$ all achieve the same product $W_2 W_1 = P_k$ and the same norms.

Problem 4

(a)

With $x_0 = 0$, we have

$$\begin{aligned} x_1 &= Bu_0, \\ x_2 &= ABu_0 + Bu_1, \\ x_3 &= A^2Bu_0 + ABu_1 + Bu_2, \\ &\vdots \\ x_k &= \sum_{i=0}^{k-1} A^{k-1-i} B\, u_i. \end{aligned}$$

Hence

$$\begin{aligned} y_k = Cx_k + Du_k &= Du_k + C \sum_{i=0}^{k-1} A^{k-1-i} B\, u_i \\ &= \sum_{\ell=0}^{k} K_\ell\, u_{k-\ell}, \end{aligned}$$

where $K_0 = D$ and $K_\ell = CA^{\ell-1}B$ for $\ell \geq 1$. With zero-padding for negative indices we have, for any $k \in \{0, \ldots, L\}$,

$$\boxed{y_k = \sum_{\ell=0}^{L} K_\ell\, u_{k-\ell}}$$

so $\{y_k\}$ is the (causal) convolution of $\{u_k\}$ with the kernel $K = \{K_\ell\}_{\ell=0}^{L}$.

(b)

(i)

Let $n = 1$ and $A = \alpha$, $B = \beta$, $C = \gamma$, $D = \delta$. Then

$$K_0 = \delta, \qquad K_\ell = \gamma\, \alpha^{\ell-1}\beta \quad (\ell \geq 1).$$

With $\alpha = 0.8,\ \beta = 1,\ \gamma = 1.5$ and $L = 4$,

$$\boxed{K_0 = \delta,\ K_1 = 1.5,\ K_2 = 1.5(0.8) = 1.2,\ K_3 = 1.5(0.8)^2 = 0.96,\ K_4 = 1.5(0.8)^3 = 0.768.}$$

This is exponentially decaying.

(ii)

$$A = \begin{bmatrix} 0.7 & 0.1 \\ 0.2 & 0.6 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0.$$

Then

$$K_0 = 0, \qquad K_1 = CB = 1, \qquad K_2 = CAB = C \begin{bmatrix} 0.7 \\ 0.2 \end{bmatrix} = 0.7.$$

Compute A^2:

$$A^2 = \begin{bmatrix} 0.7 & 0.1 \\ 0.2 & 0.6 \end{bmatrix}^2 = \begin{bmatrix} 0.51 & 0.13 \\ 0.26 & 0.38 \end{bmatrix}.$$

Hence

$$K_3 = CA^2B = C \begin{bmatrix} 0.51 \\ 0.26 \end{bmatrix} = 0.51.$$

So up to $L = 3$:

$$\boxed{K_0 = 0, \quad K_1 = 1, \quad K_2 = 0.7, \quad K_3 = 0.51.}$$

These coefficients form the impulse response $1, 0.7, 0.51, \dots$, whose decay rate is governed by the eigenvalues of A (both inside the unit disk).

(c)

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(d)

One strategy is doubling. Precompute A^{2^j} for $j = 0, \dots, \lfloor \log_2 L \rfloor$ by repeated squaring (depth $O(\log L)$). For any $\ell - 1$, expand it in binary and multiply the needed A^{2^j} blocks to apply $A^{\ell-1}$ to B. This gives all $K_\ell = C(A^{\ell-1}B)$ with work near $O(n^\omega \log L + L\,n^2)$ and depth $O(\log L)$ (here n^ω denotes matrix-multiply cost).

(e)

If $A = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$, then

$$K_\ell = CA^{\ell-1}B = C\,\mathrm{diag}(\lambda_1^{\ell-1}, \dots, \lambda_n^{\ell-1})\,B = \sum_{i=1}^{n} (Ce_i)\,(e_i^\top B)\,\lambda_i^{\ell-1}.$$

Hence K_ℓ is a sum of rank-1 exponentials (one per mode). Precompute the static factors $c_i := Ce_i$ and $b_i^\top := e_i^\top B$, and then update λ_i^ℓ via scalar recurrences $\lambda_i^\ell = \lambda_i^{\ell-1}\lambda_i$. This costs $O(nd)$ per step (or $O(ndL)$ total work) and parallelizes perfectly across i (critical path $O(\log L)$ with binary powering if desired).

(f)

Let $r := \|p\|_2$ and $q := p/r$ so that $qq^\top$ is the rank-1 projector onto $\operatorname{span}(p)$. Since I and $qq^\top$ commute and $qq^\top$ is idempotent,

$$A = I + r^2 qq^\top \quad \Longrightarrow \quad A^\ell = I + \big((1+r^2)^\ell - 1\big)\, qq^\top \qquad (\ell \geq 0).$$

Therefore, for $\ell \geq 1$,

$$\boxed{K_\ell = CA^{\ell-1}B = CB \;+\; \big((1+r^2)^{\ell-1} - 1\big)\, (Cq)\, (q^\top B).}$$

Thus the entire kernel is obtained from two precomputed factors $s_C := Cq \in \mathbb{R}$ and $s_B := q^\top B \in \mathbb{R}^{1\times d}$ plus the scalar sequence $(1+r^2)^{\ell-1}$. Each K_ℓ costs $O(d)$ time (no $n \times n$ multiplies), and the sequence can be formed in $O(Ld)$ work with $O(1)$ memory or in $O(\log L)$ depth using binary powering. Include $K_0 = D$.